Replicating and extending the analysis of Braboszcz et al. (2017) on OpenNeuro `ds003969`

October 2026

Abstract

We replicate the EEG analysis of Braboszcz et al. (2017), who reported higher gamma (60–110 Hz) power in meditators from three traditions than in controls, higher alpha power in Vipassana practitioners, and a correlation between gamma power and meditation experience. Using the public OpenNeuro dataset `ds003969`, we rebuilt their pipeline in MNE-Python, replacing manual preprocessing steps with automated equivalents. Without ICA, our per-subject parieto-occipital gamma closely matches the paper's own values ($r = 0.97$). The gamma and alpha group differences replicate, with or without automated ICA, though ICA weakens the gamma effect for Vipassana practitioners. The correlation between gamma and log practice hours reproduces exactly ($r = 0.33$), but only after removing two outliers. When outlier detection is repeated within the bootstrap, the confidence interval includes zero, and our own gamma values show no relationship. We then extend the analysis to criticality, measuring long-range temporal correlations (DFA) and functional excitation/inhibition balance (fEI). DFA shows no differences between meditators and controls, consistent with recent work. Vipassana practitioners show higher alpha fEI, which closely tracks their higher alpha power. Beta DFA rises from the first to the second meditation block, but this cannot be separated from a generic time-on-task effect. These results are robust to removing ICA and to using shorter analysis epochs.

1 Introduction

In an effort to learn more about EEG preprocessing and analysis, this project aimed to replicate and extend the findings of Braboszcz et al. (2017) using the OpenNeuro dataset `ds003969`. This paper analyses EEG recordings for meditators from 3 different traditions alongside controls, across meditation and instructed mind-wandering conditions. Analysis focused on spectral power differences and their relationship to meditation experience.

To be clear, this is a replication of the analysis and EEG preprocessing of Braboszcz et al. (2017), using the publicly available dataset `ds003969` which they released, rather than a replication of their result on different data.

In addition to replicating the original findings, this project incorporated additional analysis looking at criticality measures, inspired by Lewis-Healey et al. (2026)'s recent findings on effort(lessness) and criticality during meditation. Criticality measures used were based on the detrended fluctuation analysis (DFA) framework of Linkenkaer-Hansen et al. (2001) and Hardstone et al. (2012) with functional excitation/inhibition balance (fEI) providing another complementary measure, based on the approach of Bruining et al. (2020).

The concrete tasks were:

- **Build an equivalent pipeline in MNE-Python**, matching the authors' description of their MATLAB pipeline as closely as the text allows.
- **Replicate the headline results on spectral power:**
  - higher parieto-occipital gamma (60–110 Hz) in each meditator group than in controls, as a trait and during MED;
  - higher alpha (7–11 Hz) for Vipassana practitioners than in each of the other groups during meditation;
  - higher parieto-occipital gamma for meditators correlated with log lifetime practice hours.
- **Extend the analysis to criticality.** Compute long-range temporal correlations (DFA) and functional excitation/inhibition balance (fEI), testing the hypotheses that:
  - DFA and fEI have no trait differences between meditators and controls;
  - DFA increases over time during meditation.

2 The original paper

Braboszcz et al. (2017) recorded EEG data from experienced meditators and non-meditator controls, across meditation and instructed mind-wandering conditions. Experienced meditators came from three different meditation traditions: Himalayan Yoga, Isha Shoonya, and Vipassana.

Their analysis focuses on differences in resting state EEG spectral power in particular frequency bands between meditators and controls. Finding some significant differences, they go on to find a relationship between the difference in gamma power observed in meditators and lifetime hours of meditative practice.

Participants and setting. Data were collected at the Meditation Research Institute in Rishikesh, India. Meditators were recruited for their expertise in one of three traditions:

- **Himalayan Yoga (HYT):** silent mantra repetition, with or without breath awareness.
- **Isha Shoonya (ISY):** a practice of conscious "non-doing" with no object of focus.
- **Vipassana (VIP, S. N. Goenka style):** scanning the body from head to toes and back, observing sensations with equanimity.
- **Transcendental Meditation (TM):** a practice involving the silent repetition of a mantra, typically for 20 minutes twice a day. These were excluded from the main analysis as there were only a few participants practicing TM.

Controls (CTR) had no meditation practice. From a reported 99 people recruited (27 HYT, 20 ISY, 20 VIP, 32 CTR), the authors formed four groups of 16, matched on age and gender, for 64 in total. Mean ages were 40–49 per group and did not differ significantly. HYT and VIP had more lifetime practice than ISY (means of roughly 15,000 and 9,000 hours against 2,600).

Design. Each participant did two 20-minute blocks, with eyes closed in a dark, soundproof room:

- **Meditation (MED):** 10 minutes of breath focus, then 10 minutes of the participant's own practice, cued over an intercom. Controls continued breath focus for the second half.
- **Instructed mind-wandering (IMW):** recalling autobiographical events from childhood to the recent past, avoiding emotionally charged memories.

Block order was counterbalanced within each group, and participants filled in questionnaires between the two blocks. Auditory and visual tasks were also run in the sessions; their results are not reported in the paper. EEG was recorded with a 64-channel Biosemi ActiveTwo system at 1024 Hz, with mastoid and EOG electrodes. The analysis compared the last 10 minutes of MED with the last 10 minutes of IMW.

Analysis. Recordings were referenced to the right mastoid, downsampled to 256 Hz and high-pass filtered at 1 Hz. Artefactual segments were rejected with EEGLAB's `pop_rejcont`, bad channels were removed by eye, and 1–5 eye and temporal-muscle ICA components per subject were removed after visual inspection. Spectra were computed from 1 s epochs and compared with cluster permutation tests over electrodes and Welch or mixed ANOVAs.

Findings.

- **Gamma (60–110 Hz):** all three meditator groups had higher parieto-occipital power than controls, both as a trait (MED and IMW combined) and during MED.
- **Alpha (7–11 Hz):** VIP had more power than each of the other three groups, during MED and IMW.
- **Experience:** parieto-occipital gamma correlated with log lifetime practice hours ($r = 0.33$; their Fig. 4D).
- **Artefact controls:** temporal-muscle components did not correlate with gamma. Eye-movement components did ($r = 0.33$), but they did not correlate with experience, and the authors concluded that artefacts were unlikely to explain the gamma findings.

The paper's spectra by group are reproduced in Fig. 8.

3 Data

The dataset. OpenNeuro `ds003969` (v1.0.0) contains 64-channel Biosemi recordings of 98 participants: 31 CTR, 24 HYT ("HTR" in the dataset), 20 ISY ("SNY"), 19 VIP and 4 Transcendental Meditation. These counts don't line up exactly with those reported in the paper. Each participant's two sessions are recorded in four ∼10-minute BDF files: `think1`, `think2`, `med1breath` (breath focus) and `med2` (the tradition's own practice; breath focus for controls). The paper's analysis periods are therefore `think2` (IMW) and `med2` (MED). Due to a defect, electrode `Fp1` is replaced by `EXG8`, the spare electrode wired in its place.

Session timing. The recording start times in the BDF headers show how the blocks were scheduled. Both blocks of a session were recorded back to back, and the two sessions were separated by a median of 66 minutes (range 47–89), all on one day. Roughly half of the participants did the MED session first (51 of 98) and half the IMW session first (47), as recorded in the `first_session` column of `participants.tsv`; this agrees with the timestamps for all 98. Within a session the order is fixed, so `med2` always directly follows `med1breath`.

The paper's subject list (Zenodo). The dataset does not record which 64 participants the paper analysed. The authors' earlier Zenodo deposit (doi 10.5281/zenodo.57911) lists them by lab ID, with their per-subject median parieto-occipital gamma and alpha values. We mapped these lab IDs to `ds003969` subjects using the recording dates in the BDF headers, cross-checked against age. 63 of the 64 are in `ds003969`; HYT26's recording is missing from the publicly released data. All results in this report use this matched sample (16 CTR, 15 HYT, 16 ISY, 16 VIP, 252 recordings). The paper's per-subject values are also used directly, to check the replication subject by subject.

4 Methodology

4.1 Pipeline replication

Table 1 sets the paper's description against what was implemented. Every step was either matched as closely as the text allows. In cases where manual decisions were made for e.g. artefact or bad channel detection, these were replaced by an automated equivalent. Initially we attempted to reconstruct these manual judgements on bad channels and bad ICA components, but it quickly became clear that we were not able to make the same manual judgements reliably.

Counts in the table are for the 252 matched-sample recordings.

Two areas created significant differences in the processed data: the sample selection and manual decisions in the ICA and bad channel and segment procedures. Efforts were made during the project to match the authors' decisions as closely as possible. However, this was not entirely possible and discrepancies remained.

Sample. The project was initially run on all 94 non-TM participants, because the dataset does not say which participants the paper analysed. That sample is not the paper's: it has about 50% more participants, is not age-matched, and its additional controls are younger than the paper's. Its group comparisons therefore do not test the paper's claims, and they gave a misleading picture of how well the paper replicates. Recovering the paper's subject list from its Zenodo deposit (Section 3) fixed this, and all results here are from the sample restricted to that list. However, as noted above, this sample is still not exactly the same as the paper's, as participant HYT26 (as they are labelled in the Zenodo deposit) has no data present in the public dataset.

ICA. The paper's component choices were made by eye and cannot be recovered, so several automated rules were tried while developing the pipeline:

- An uncapped rule (muscle threshold 0.5, EOG z>3) often removed more than 10 of 20 components per recording, far more than the paper's 1–5.
- Raising the thresholds too far (muscle 1.0, EOG z>5) removed essentially nothing.
- An intermediate uncapped rule (muscle 0.75, EOG z>3) removed a median of about 2 but up to 17.
- Caps were adopted to mirror the counts in the paper: at least one muscle component, at most four EOG and at most five in total.

As discrepancies remained even with the closest matching automated approach, the final analysis reports results from both the capped rule and a no-ICA pipeline to provide a more complete picture.

Other manual decisions. Beyond ICA component removal, the paper reports a number of other manual decisions which couldn't be replicated exactly:

- Manual selection of bad channels from EEG recordings.
- Manual selection which recordings to apply a harsher segment rejection procedure to.

Again, these decisions could not be exactly replicated, and may contribute to differences between this replication and the original paper's results.

4.2 Spectral analysis

Here we tried to replicate Braboszcz et al.'s approach as closely as possible, but have added extensions where it seems their approach was lacking.

Step	Paper	This replication
Sample	16 per group, matched on age and gender from a larger pool; TM not included	The paper's matched list, 63 of 64 available (See section 3)
Reference	Right mastoid	Right mastoid (`EXG6`)
Resample, filter	256 Hz; 1 Hz IIR high-pass, order 6, 0.3 Hz transition	Same
Segment rejection	EEGLAB `pop_rejcont`: 1 s epochs, 0.5 s overlap, 0–10 Hz >17 dB, 35–128 Hz >14 dB, contiguous 8; thresholds "in rare cases" adjusted by 1–5 dB after visual review	Line-by-line Python port with unit tests, including EEGLAB quirks such as `contiguous` acting as a merge distance. Underspecified manual threshold adjustment not reproduced.
Bad channels	Manual; 0–18 per subject, mean ≈5; interpolation not stated	Automated: union of MNE waveform LOF (thr. 1.5) and a custom spectral LOF on 1–45 Hz log-PSD (thr. 2). Capped at 20 per recording, dropping the weakest suggestions first. Mean 6.7 per recording, median 6, range 0–20. By default these are excluded and cluster means use the remaining channels, but for the electrode-level cluster tests they are interpolated with spherical splines as all channels are required
ICA	Infomax; 1–5 eye/temporal-muscle components chosen visually by one experimenter	Infomax with 20 components (count not given in the paper), fit on good EEG channels. Components chosen automatically with `mne-python` functions `find_bads_muscle` (thr. 0.75) and `find_bads_eog` (z>3), capped at ≥1 muscle, ≤4 EOG and ≤5 in total: mean 3.3 removed (2.4 muscle, 0.9 EOG), range 1–5. A second pipeline with no ICA is reported alongside
Spectra	1 s non-overlapping epochs, Hanning taper, FFT, $10\log_{10}$; 2–110 Hz, 50 Hz band excluded	Same epoching and taper (Welch with one segment per epoch); epochs overlapping `BAD` spans dropped; linear power averaged over epochs, then converted to dB
Statistics	FieldTrip cluster permutation on log power (2000 resamples) over electrodes; Welch/mixed ANOVA	Cluster permutation over electrodes (2000 permutations; Appendix A); Welch t-tests on the mean over an 8-electrode parieto-occipital cluster (Section 4.2)

Table 1: Paper pipeline compared with the replication.

- **Parieto-occipital cluster differences.** Power in each band was averaged over an 8-electrode parieto-occipital cluster (O1, Oz, O2, PO7, PO3, POz, PO4, PO8), excluding the recording's bad channels, and expressed in dB. Bands follow the paper: gamma 60–110 Hz and alpha 7–11 Hz.
- **Cluster-based permutation tests.** Cluster based permutation tests were used to assess the significance of any spectral differences, following the paper's approach. An FDR correction was then applied to these p-values, which the original paper did not apply.
- **Conditions.** MED is `med2`, IMW is `think2`, and the trait measure is each subject's mean of the two.
- **Comparison with the paper's values.** The paper's own per-subject gamma power values are available on Zenodo. These were compared to the values calculated here to check for consistency of the two pipelines.

Two minor points came up along the way:

- The paper's y-axis (∼45 dB at low frequencies) is offset from a correctly scaled PSD in $\mu V^2/Hz$ by a constant of roughly 35–45 dB. This is consistent with an unnormalised squared FFT magnitude. Because it is a constant offset in dB, between-group differences are unaffected.
- The 100 Hz line-noise harmonic falls inside the gamma band and is not excluded. However, visual inspection showed nothing abnormal at 100 Hz.

4.3 Criticality analysis

Measures. The Hilbert amplitude envelope was computed in alpha (8–13 Hz), beta (13–30 Hz), and gamma (60–110 Hz). These bands were chosen to better match Lewis-Healey et al. (2026), rather than Braboszcz et al. (2017).

DFA and fEI used `crosci` on non-overlapping 180 s epochs, with DFA fitted over 5–30 s and fEI computed in 5 s windows with 80% overlap. Values were averaged over epochs and good channels to give one value

per subject, task and band. Manual implementations of DFA and fEI were first written and checked against `crosci`.

These settings follow Lewis-Healey et al. (2026): `crosci`, the same band definitions, 3-minute windows, DFA up to 30 s with 50% overlap between DFA windows, fEI with 5 s windows at 80% overlap, and fEI only where DFA > 0.6. That study differs in a few respects:

- Band-specific minimum DFA windows (alpha 3.98 s, beta 2.24 s).
- The prior preprocessing pipeline: average reference, 0.5–45 Hz FIR filtering, FastICA with manual component rejection.
- Rejection on 5 s epochs, instead of using `pop_rejcont`.

Hypotheses. A number of hypotheses were formed and tested, based on the results in Lewis-Healey et al. (2026):

- H_1: There is no trait difference in criticality measures (DFA and fEI) between groups.
- H_2: Meditators show an increase in DFA criticality over time whilst meditating, i.e. from `med1breath` to `med2`.

Given the results in Lewis-Healey et al. (2026), we also speculated that any differences would be most likely to manifest in alpha or beta. Gamma was included in the analysis for completeness, given the significant gamma power differences reported by Braboszcz et al. (2017).

Data loss. An epoch is dropped if *any* `BAD` span overlaps it. Even a 2 s `pop_rejcont` flag therefore discards 180 s of data. As a consequence:

- Only 149 of 252 recordings (59%) have at least one usable epoch, from 55 of the 63 subjects.
- `think2` retains only 28 of 63 recordings.
- Usable recordings average 1.7 epochs, about 5 of 10 minutes.
- Only 26 meditators and 6 controls have both meditation blocks.

This data loss meant that the different groups were no longer balanced in terms of having equal numbers doing the meditation vs mind wandering tasks first. To stop this influencing results an indicator variable was included in all regressions for whether a recording was in the second group of tasks. If ordering was important then this variable should pick up this effect, rather than it confounding the main results.

This was the most important practical weakness of the criticality analysis. An epoch-length sweep on the same recordings shows how much shorter epochs would recover: 95% of recordings keep at least one epoch at 60 s and 88% at 90 s, against 59% at 180 s. Analysis was repeated on 90 s epochs to assess the impact of shorter epoch lengths on results.

5 Results

5.1 Spectral power

Agreement with the paper's per-subject values. Without ICA, this replication's parieto-occipital gamma matches the paper's own per-subject values closely: Pearson $r = 0.97$ in MED, 0.94 in IMW and 0.96 for the trait measure, with a residual scatter of 1.2–1.6 dB around a constant offset (Fig. 1a). Alpha agrees at $r = 0.94$–0.95. With the automated ICA pipeline, agreement on gamma falls to $r = 0.68$ in MED and 0.75 for the trait measure (Fig. 1b), while alpha is almost unaffected ($r = 0.90$–0.93).

The ICA step therefore accounts for most of the remaining disagreement on gamma, and the no-ICA pipeline is the closer reproduction of the paper's processing. For this reason key results are reported both with and without ICA processing, but are reassuringly found to be robust to these differences. The difference in gamma values can be seen in figure 1b to be concentrated in a few subjects, and a further analysis of this work would likely focus on these specific subjects to understand the sources of discrepancy.

Group differences on the parieto-occipital cluster. Fig. 2 gives the group differences on the 8-electrode cluster mean, with Welch t-tests, for the paper's values and for both pipelines; Table 4 (Appendix D) lists the values.

- **Gamma, no ICA.** Each meditator group exceeds controls by 3.5–4.8 dB in MED and as a trait, all with $p \leq .02$. These are within 1 dB of the paper's own values for every contrast.
- **Gamma, automated ICA.** The differences shrink by about 1 dB on average. VIP is affected most (MED: +2.6 dB, $p = .12$; trait: +2.4 dB, $p = .09$), while HYT and ISY stay significant. Pooled meditators still exceed controls in MED (+3.1 dB, $p = .011$) and as a trait (+3.3 dB, $p = .002$).

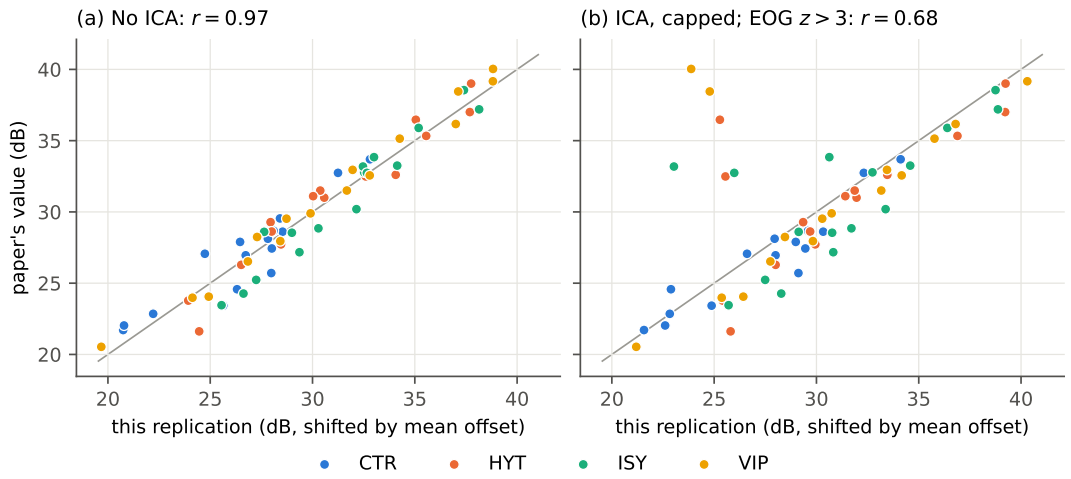


Figure 1: Per-subject parieto-occipital gamma in MED: this replication against the paper's values from Zenodo (63 subjects). This replication's values are shifted by the mean offset between the two scales.

- **Alpha.** VIP has more alpha than controls in MED in all three sources (3.5–4.3 dB, $p \leq .014$), and more than HYT and ISY combined (about 4 dB, $p \leq .007$). The effect is weaker in IMW ($p = .03$–$.07$ in this replication).

So this replication largely confirms the paper's findings in these areas, with some attenuation of estimates when using the automated ICA pipeline.

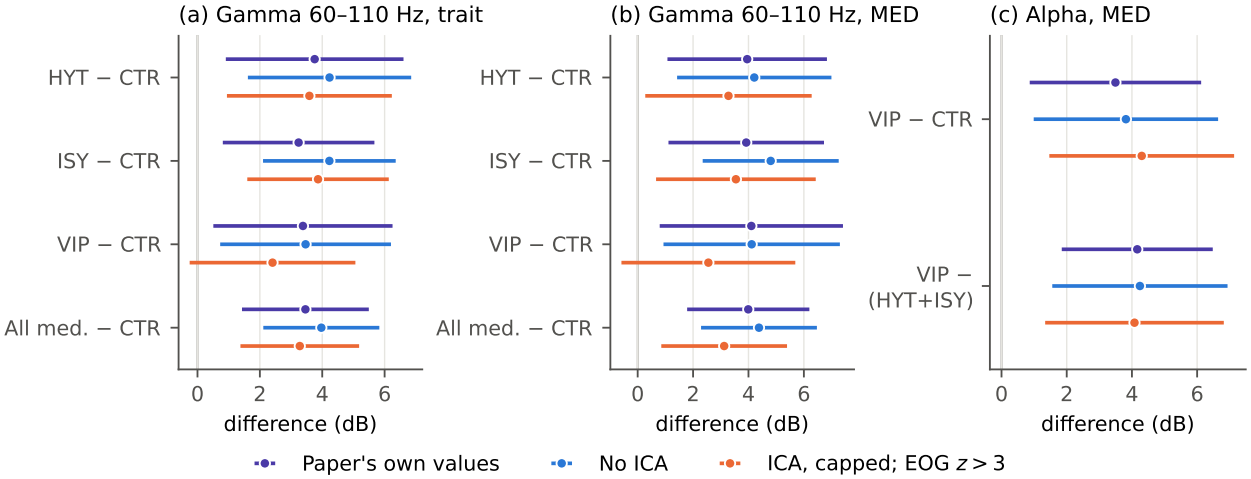


Figure 2: Group differences on the parieto-occipital cluster with 95% confidence intervals (Welch), for the paper's values and both pipelines.

Cluster permutation tests. Table 2 gives the smallest cluster p-value for each electrode-level test, following the paper's procedure (Appendix A, where the topographies are shown). All clusters with $p < .05$ are positive: meditators above controls in gamma, VIP above the other groups in alpha.

- **Gamma, no ICA.** All three groups have a significant cluster against controls in MED ($p = .017$, $.024$, $.042$ for HYT, ISY, VIP). As a trait, HYT and ISY do ($p = .029$, $.032$) and VIP narrowly misses ($p = .055$). The clusters are centred on parieto-occipital electrodes; for HYT and the pooled meditators in MED they extend over most of the scalp (43 electrodes).
- **Gamma, automated ICA.** Only ISY keeps a significant MED cluster ($p = .042$, 9 parieto-occipital electrodes). HYT and ISY remain significant as a trait and in IMW, and VIP has no cluster below $p = .13$ in any condition.
- **Alpha.** In MED, VIP exceeds controls, HYT and ISY in both pipelines ($p = .013$–$.040$), with broad clusters of 46–59 electrodes. In IMW, VIP exceeds HYT in both pipelines and exceeds controls only with ICA ($p = .023$).

- **MED vs IMW within groups.** No group shows a significant difference between conditions in either band (smallest $p = .12$).

So the cluster permutation tests largely replicate the paper's findings, with some differences depending on the pipeline used, but nothing that fundamentally contradicts the original findings.

		No ICA			Capped ICA		
Band	Contrast	MED	IMW	trait	MED	IMW	trait
Gamma	HYT − CTR	**.017** (43)	**.042** (23)	**.029** (40)	.066 (5)	**.038** (26)	**.034** (26)
	ISY − CTR	**.024** (32)	.059 (9)	**.032** (22)	**.042** (9)	**.035** (21)	**.032** (18)
	VIP − CTR	**.042** (19)	.114 (4)	.055 (11)	.179 (1)	.190 (1)	.136 (1)
	All meditators − CTR	**.021** (43)	**.040** (19)	**.033** (36)	.053 (10)	**.034** (28)	**.041** (22)
Alpha	VIP − CTR	**.031** (48)	.065 (14)	**.034** (43)	**.013** (59)	**.023** (54)	**.017** (59)
	VIP − HYT	**.022** (59)	**.029** (45)	**.025** (54)	**.033** (47)	**.033** (44)	**.034** (46)
	VIP − ISY	**.038** (46)	.076 (8)	.056 (28)	**.040** (46)	–	.089 (4)
	All meditators − CTR	–	–	–	–	–	–

Table 2: Cluster permutation tests over electrodes: smallest cluster p-value, with the number of electrodes in that cluster in brackets; $p < .05$ in bold; – means no electrode passed the cluster-forming threshold. Two-tailed, 2000 permutations, 63 subjects.

Correcting for the number of tests. The paper's cluster tests correct across electrodes within each test, but not across the many tests run. To see whether this matters, we applied a Benjamini–Hochberg false discovery rate (FDR) correction (Benjamini and Hochberg, 1995) at 5% to each test's smallest cluster p-value, within each pipeline. The family of tests is the 23 tests that correspond to claims in the paper: gamma, each tradition against controls in MED, IMW and trait; alpha, VIP against each other group in MED and IMW; MED against IMW within each group in both bands.

- In each pipeline, 10 of the paper's 23 tests reach $p < .05$, against about 1 expected if there were no effects at all.
- None survive the FDR correction, in either pipeline. The smallest adjusted p-value is $q = .097$ over the paper's tests in both pipelines, and .106 (no ICA) and .114 (capped ICA) when all 38 tests run here are included.
- The reason is that every significant cluster p-value lies between .013 and .042. Benjamini–Hochberg needs the k-th smallest of m p-values to be at most $0.05\,k/m$; with 10 hits out of 23, the 10th would need to be below .022.

However, a 10% FDR correction would keep all of them, so the results are clearly not entirely spurious. A reasonable conclusion is that the paper's reported p-values slightly overstate their statistical significance, but their main conclusions are not entirely undermined.

Gamma and meditation experience. The paper reports a skipped correlation of $r = 0.33$ (95% CI [0.06, 0.53]) between meditators' trait parieto-occipital gamma and log lifetime practice hours (its Fig. 4D). We reimplemented the skipped correlation (Pernet, Wilcox and Rousselet, 2013): bivariate outliers are detected with the projection method and removed, and Pearson's r is computed on the rest. Results are in Table 3 and Fig. 3.

- **The point estimate reproduces exactly.** On the paper's values for its 48 meditators, two outliers are removed (HYT7, with many hours and low gamma; ISY6, with few hours and high gamma), giving $r = 0.33$. Without the outlier step the correlation is weak: Pearson $r = 0.16$ ($p = .27$).
- **The confidence interval depends on how the bootstrap treats the outliers.** If the two outliers are removed once and Pearson's r is bootstrapped on the remaining 46 meditators, the 95% CI is [0.08, 0.54], matching the paper's up to monte carlo error. This is presumably what the paper did. It ignores the uncertainty in which points get flagged as outliers. Redoing the outlier detection within each bootstrap resample gives [−0.14, 0.51], which includes zero.
- **The estimate is fragile.** Dropping HYT26, the one meditator missing from `ds003969`, changes which points are flagged as outliers and lowers r to 0.24.
- **This replication's own gamma shows no clear relationship:** $r = 0.16$ without ICA (no outliers flagged) and $r = -0.05$ with capped ICA. Both CIs include zero under either bootstrap approach.

Overall this suggests that the reported correlation between trait parieto-occipital gamma and meditation experience is extremely sensitive to outlier handling. Visually, considering figure 3a, which is an exact

replication of the paper's figure 4D, it's clear that the slope of the fitted line is driven by the choice to remove two outliers.

					95% CI, outliers...	
Gamma values	n	Pearson r	Skipped r	Outliers	removed once	redetected
Paper's values	48	0.16	0.33	HYT7 ISY6	[0.08, 0.54]	[−0.14, 0.51]
without HYT26	47	0.17	0.24	HYT7	[−0.07, 0.50]	[−0.17, 0.50]
No ICA	47	0.16	0.16	none	[−0.16, 0.45]	[−0.19, 0.51]
Capped ICA	47	0.06	−0.05	HYT20	[−0.36, 0.27]	[−0.34, 0.34]
Paper (Fig. 4D)	48		0.33		[0.06, 0.53]	

Table 3: Correlation between trait parieto-occipital gamma and $\log_{10}$ lifetime practice hours among meditators. Skipped correlations with percentile bootstrap CIs (2000 resamples). "Removed once" bootstraps Pearson's r on the subjects left after removing the outliers found in the full sample; "redetected" repeats the outlier detection in each resample. The upper rows use the paper's values, for all 48 meditators and for the 47 in `ds003969`; the lower rows use this replication's values. Practice hours are from the paper's Zenodo file.

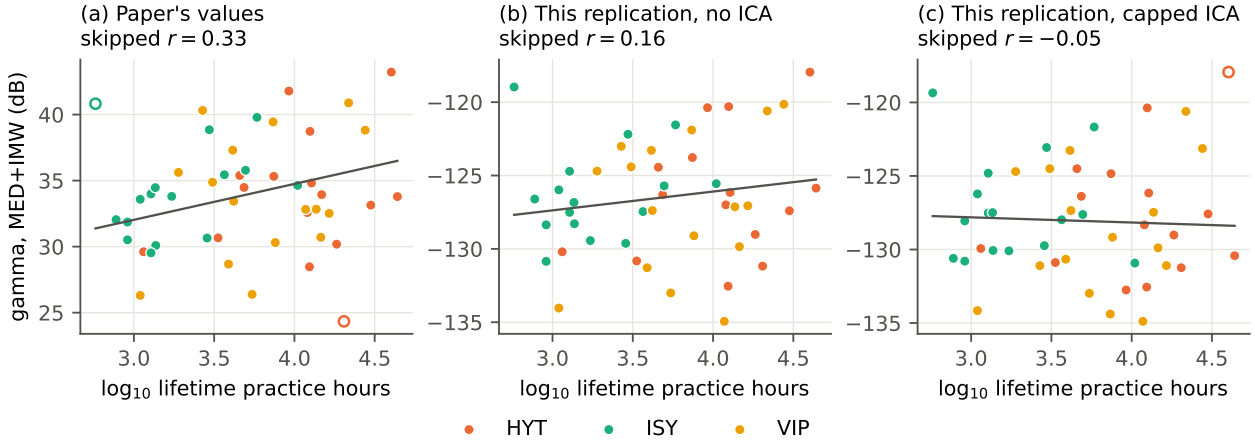


Figure 3: Trait parieto-occipital gamma against lifetime practice hours (the paper's Fig. 4D). Open circles are the outliers removed by the skipped correlation; lines are fitted to the remaining points.

5.2 Criticality

The main reported results here are for the pipeline that uses automated ICA component removal. All analysis was also run with the no-ICA pipeline and the results were qualitatively similar, with close point estimates and no major differences in significance.

Mean DFA exponents were above 0.5 in every band (0.74 in alpha, 0.69 in beta and 0.80 in gamma), indicating long-range temporal correlations in the amplitude envelopes. Mean fEI was below 1 in every band (0.93 in alpha, 0.85 in beta and 0.91 in gamma), which in the fEI framework indicates slightly inhibition-dominated dynamics. Hypotheses were tested using OLS regressions of the subject-level DFA exponent (and, for H_1, fEI) with standard errors clustered by subject. In each model, we also include an indicator variable for being in the second pair of tasks as a control for reasons outlined in section 4.3. Fig. 4 shows the key terms, and Tables 5 and 6 (Appendix D) give the full estimates.

H_1: no trait difference between groups. The trait measure uses `med2` and `think2`, as in the spectral analysis: 72 recordings from 49 subjects.

- **DFA.** No meditator group differs from controls in any band (all $p \geq .15$). A joint test that all four groups are equal gives $p = .74$ (alpha), .37 (beta) and .86 (gamma). This is consistent with H_1, and with Lewis-Healey et al. (2026). The confidence intervals are wide, though (±0.07–0.10 in alpha and beta, ±0.11–0.16 in gamma), so moderate group differences cannot be ruled out.
- **fEI.** Beta and gamma show no group differences (joint $p = .62$ and .85). In alpha, VIP has higher fEI than controls (+0.20, $p = .003$), and the joint test rejects equal groups ($p = .003$), so H_1 does not hold for alpha fEI. HYT and ISY do not differ from controls. This was unexpected but in ways aligns with the

finding of higher alpha power in VIP. This result does not appear to be driven by outliers: alpha fEI lies above the control median for 11 of the 12 VIP subjects.

Interestingly, this fEI result mirrors the alpha power results: VIP shows higher alpha power than controls, while HYT and ISY do not differ from controls. Measuring alpha power in the same band as fEI (8–13 Hz) we find a correlation of 0.59 ($p < .001$) between alpha power and alpha fEI across subjects. This is an interesting relationship, and two explanations that cannot be distinguished by this data present themselves: it could be that higher alpha power drives observed increases in fEI (fEI estimates may depend on the prominence of the oscillation in the data), or that both are driven by a common underlying factor such as excitation/inhibition balance.

H_2: DFA increases during meditation. This uses all four blocks for the meditators: 117 recordings from 40 subjects.

- **Beta DFA rises from `med1breath` to `med2`** (+0.061, $p = .009$). Alpha shows a smaller and less significant rise (+0.039, $p = .055$), and gamma none.
- **This rise cannot be separated from time on task.** `med2` always directly follows `med1breath`, so the first-to-second-block change also contains any generic effect of time spent in the task. The comparison with the thinking pair, (`med2` − `med1breath`) − (`think2` − `think1`), isolates the part specific to meditation. It has a similar point estimate and is positive in beta (+0.052) but is not significant in any band (beta $p = .15$, alpha $p = .63$, gamma $p = .97$).

So the data show an increase in beta DFA over the meditation session, in the direction H_2 predicts. However, there is insufficient evidence to conclude that the increase is specifically due to meditation rather than a generic time on task effect.

Session order. DFA is lower in the second pair of blocks, after the break, in every band. The difference is significant in beta (−0.052, $p = .044$ in the H_1 model; −0.043, $p = .014$ in the H_2 model) and close to significance in alpha ($p = .09$ in both). Subjects completed a questionnaire and a set of auditory and visual tasks between the first and second pairs of blocks. Braboszcz et al. (2017) mention these tasks but do not report their results. It's possible these tasks were responsible for the observed session order effect.

Without ICA. Appendix Fig. 7 repeats both analyses on the no-ICA pipeline. The results do not differ: no DFA estimate changes by more than 0.02 and no fEI estimate by more than 0.04. The only term that changes significance is VIP − CTR in beta fEI (+0.09, $p = .043$ without ICA; +0.05, $p = .25$ with ICA); the alpha fEI difference holds in both ($p = .005$ without ICA).

As noted in section 4.3, the choice to use base 180 second epochs for the criticality analysis means a lot of data is dropped, with only 59% of recordings having at least one usable epoch. To test the robustness of these results to that choice, we reran this analysis using 90 second epochs. This resulted in 88% of recordings having at least one usable epoch. The main results above (No group differences in DFA, higher trait fEI for VIP practitioners, and a positive change in beta DFA over the meditation session) all survived this change.

6 Conclusion

Here we have largely replicated the data processing pipeline and analysis of Braboszcz et al. (2017). As a learning exercise this has been very fruitful, covering the early stages of EEG data processing through to the final statistical analyses.

In constructing the pipeline, the main difficulty in replicating the original design was steps that required unspecified manual intervention, such as artifact rejection and channel selection. Ultimately it was not possible to exactly replicate the processing and fortunately it was not necessary, as all analysis shown above was largely robust to different choices made during preprocessing. Really this is a better outcome than an exact replication as we actually learn something about the robustness of the original findings.

The authors' (Braboszcz et al., 2017) results on gamma power in meditators and alpha power specifically in vipassana practitioners both replicate and do so robustly to different choices made in the preprocessing pipeline. However, their result on the relationship between gamma power and lifetime hours of meditative practice has been shown to be extremely fragile, depending entirely on the choice to drop two observations as outliers.

A further analysis of criticality metrics (DFA and fEI) found no trait differences in DFA between controls or meditators of any type, consistent with the findings of Lewis-Healey et al. (2026). However, there is some evidence for higher trait alpha fEI specifically in vipassana practitioners. This aligns with the finding of higher alpha power for this group, and the two findings are difficult to separate. There was some suggestive

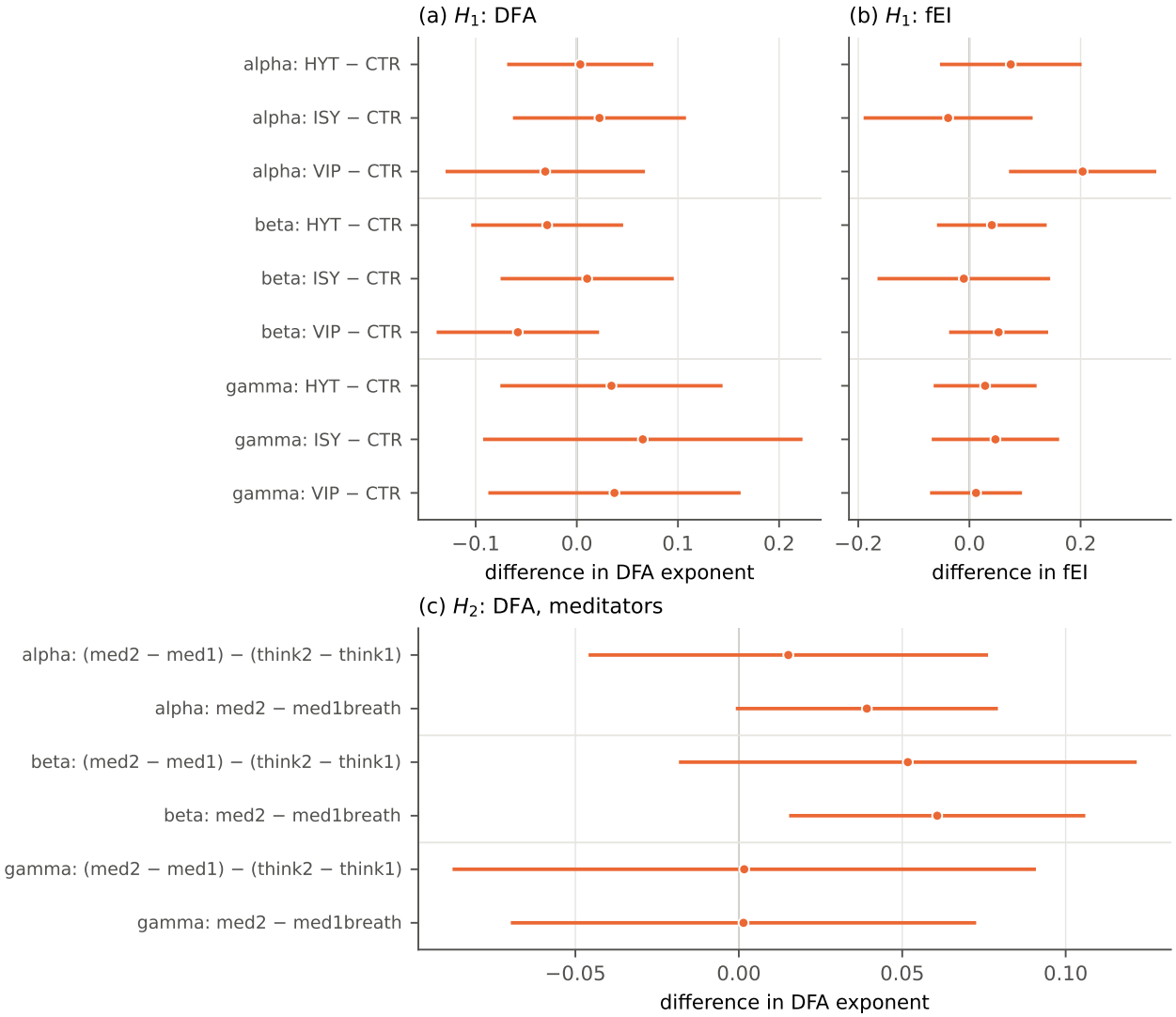


Figure 4: Key terms from the criticality regressions (capped ICA), with 95% confidence intervals. (a, b) Each meditator group against controls, `med2` and `think2`, for DFA and fEI. (c) Within meditators: the change from `med1breath` to `med2`, and that change minus the change from `think1` to `think2`.

evidence for criticality increasing for meditators during meditation, although the estimation procedure could not distinguish whether this was an effect of meditation or a more generic time on task effect.

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A Cluster permutation tests

The electrode-level tests follow the paper's description of its FieldTrip analysis:

- **Power per electrode.** Median over 1 Hz bins of $10 \log_{10}$ power in the band. Bad channels were interpolated with spherical splines so that every subject has a value at every electrode.
- **Statistic.** Student's t at each electrode: independent samples for group comparisons, paired for MED against IMW within a group.
- **Clusters.** Electrodes with $|t|$ above the two-tailed $p < .05$ threshold are grouped with neighbouring electrodes (Delaunay triangulation of the 64-channel layout). Each cluster is scored by its summed t.
- **Inference.** Group labels were shuffled 2000 times (sign flips for the paired tests), recording the largest cluster score each time. A cluster's p-value is the proportion of permutations whose largest score is at least as large. This controls the family-wise error rate across electrodes within each test, but not across tests.

Figures 5 and 6 show the group differences, with the electrodes in clusters with $p < .05$ marked. The code is in `code/cluster_stats.py` and `scripts/cluster_tests.py`, and the results are in `results/matched/cluster_tests.csv` and, with FDR-adjusted values, `cluster_tests_fdr.csv`.

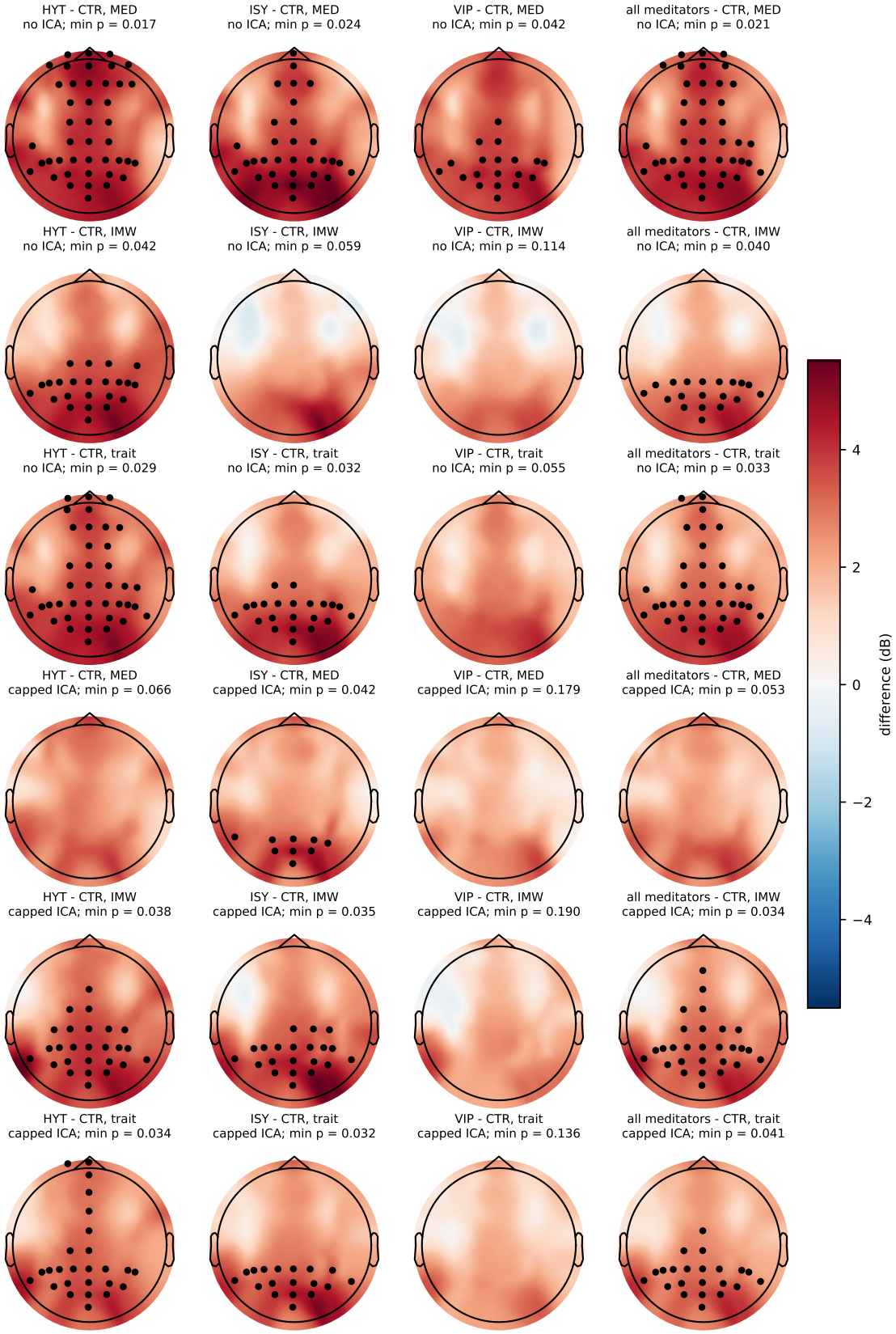


Figure 5: Gamma (60–110 Hz): mean difference from controls (dB). Black dots mark electrodes in clusters with $p < .05$. Rows 1–3: no ICA; rows 4–6: capped ICA.

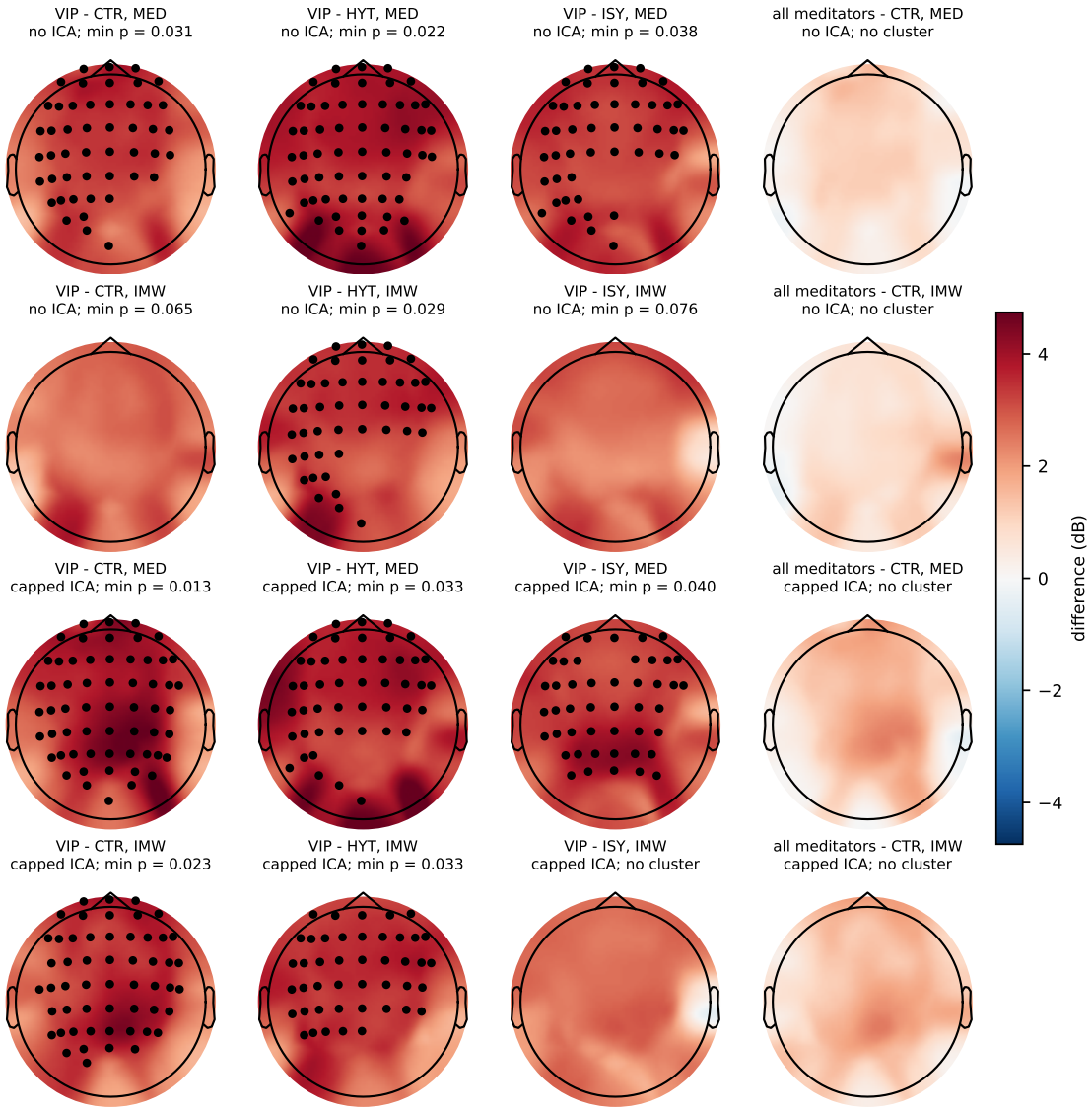


Figure 6: Alpha (7–11 Hz): VIP compared with each other group, and all meditators against controls (dB). Black dots mark electrodes in clusters with $p < .05$. Rows 1–2: no ICA; rows 3–4: capped ICA.

B Criticality without ICA

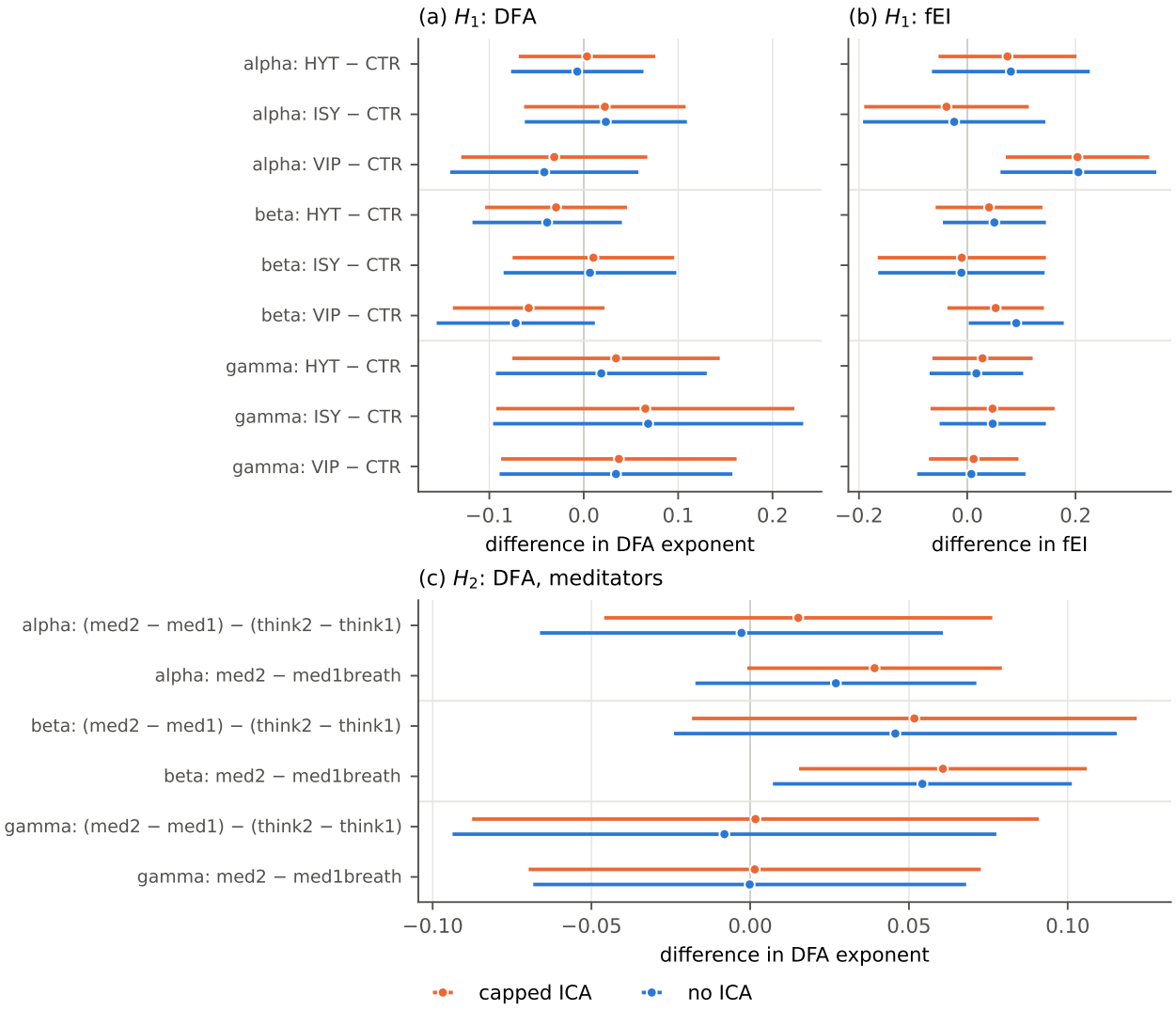


Figure 7: The terms in Fig. 4 for the capped-ICA pipeline (orange) and the no-ICA pipeline (blue), with 95% confidence intervals.

C The paper's spectra

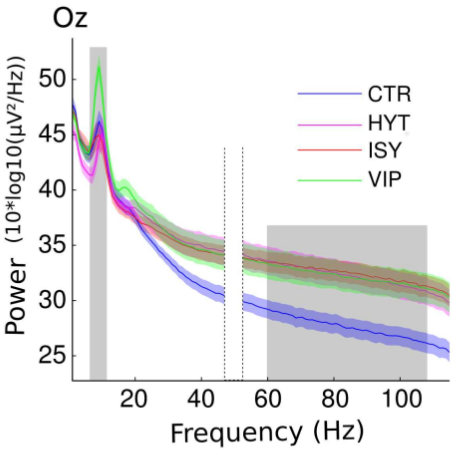


Fig 3. Spectral decomposition for occipital electrode Oz during the meditation condition in all 4 groups of subjects (CTR is the Control group; HYT is the Himalayan Yoga Tradition group; ISY is the Isha Shoonya Yoga group; VIP is the Vipassana group). Shaded regions indicate the standard error of the mean. The grey shaded area indicates the region of statistical differences between the groups after correction for multiple comparisons (see Material and Methods). The region around 50 Hz has been blanked out on the plot as it corresponds to line noise and has been excluded from analysis.

doi:10.1371/journal.pone.0170647.g003

Figure 8: Braboszcz et al. (2017), Fig. 3: Oz spectra by group (CC BY 4.0).

D Full results tables

Band	Condition	Contrast	Paper's values	No ICA	Capped ICA
Gamma 60–110 Hz	MED	HYT − CTR	+3.95 (**.012**)	+4.21 (**.006**)	+3.28 (**.042**)
		ISY − CTR	+3.92 (**.011**)	+4.80 (<**.001**)	+3.55 (**.022**)
		VIP − CTR	+4.10 (**.023**)	+4.12 (**.018**)	+2.55 (.123)
		All meditators − CTR	+3.99 (**.001**)	+4.38 (<**.001**)	+3.12 (**.011**)
	IMW	HYT − CTR	+3.56 (**.035**)	+4.26 (**.006**)	+3.90 (**.008**)
		ISY − CTR	+2.57 (.076)	+3.66 (**.005**)	+4.18 (**.001**)
		VIP − CTR	+2.66 (.071)	+2.82 (**.048**)	+2.26 (.110)
		All meditators − CTR	+2.93 (**.013**)	+3.56 (**.001**)	+3.44 (**.001**)
	trait	HYT − CTR	+3.76 (**.016**)	+4.23 (**.004**)	+3.59 (**.013**)
		ISY − CTR	+3.24 (**.014**)	+4.23 (<**.001**)	+3.87 (**.002**)
		VIP − CTR	+3.38 (**.029**)	+3.47 (**.020**)	+2.41 (.087)
		All meditators − CTR	+3.46 (**.002**)	+3.97 (<**.001**)	+3.28 (**.002**)
Alpha 7–11 Hz	MED	VIP − CTR	+3.49 (**.014**)	+3.82 (**.013**)	+4.30 (**.006**)
		VIP − (HYT+ISY)	+4.16 (**.001**)	+4.24 (**.004**)	+4.08 (**.007**)
	IMW	VIP − CTR	+2.72 (**.047**)	+2.85 (.058)	+3.07 (.056)
		VIP − (HYT+ISY)	+3.43 (**.008**)	+3.23 (**.029**)	+2.79 (.071)

Table 4: Group differences in mean power over the 8-electrode parieto-occipital cluster (dB), with Welch t-test p-values in brackets; $p < .05$ in bold. "Paper's values" puts the paper's per-subject values from Zenodo (Braboszcz et al., 2016) through the same tests. Trait is each subject's mean of MED and IMW.

	Alpha		Beta		Gamma	
Term	Estimate (SE)	p	Estimate (SE)	p	Estimate (SE)	p
DFA exponent						
HYT − CTR	+0.003 (0.037)	.925	−0.029 (0.038)	.445	+0.034 (0.056)	.541
ISY − CTR	+0.022 (0.044)	.608	+0.010 (0.044)	.815	+0.065 (0.081)	.418
VIP − CTR	−0.031 (0.050)	.535	−0.058 (0.041)	.155	+0.037 (0.064)	.558
Second pair	−0.045 (0.026)	.089	−0.052 (0.026)	**.044**	−0.022 (0.030)	.475
All groups equal	$\chi^2_3 = 1.25$	.742	$\chi^2_3 = 3.12$	.373	$\chi^2_3 = 0.75$	.861
Observations (subjects)	72 (49)		72 (49)		72 (49)	
fEI						
HYT − CTR	+0.074 (0.065)	.253	+0.040 (0.050)	.424	+0.028 (0.047)	.550
ISY − CTR	−0.038 (0.078)	.621	−0.010 (0.079)	.899	+0.047 (0.058)	.424
VIP − CTR	+0.204 (0.068)	**.003**	+0.053 (0.045)	.248	+0.012 (0.042)	.778
Second pair	−0.031 (0.033)	.352	−0.029 (0.031)	.357	+0.030 (0.034)	.384
All groups equal	$\chi^2_3 = 13.67$	**.003**	$\chi^2_3 = 1.79$	.617	$\chi^2_3 = 0.80$	.849
Observations (subjects)	72 (49)		72 (49)		72 (49)	

Table 5: H_1: OLS of the DFA exponent and of fEI on group and the second-pair indicator, using `med2` and `think2`, with standard errors clustered by subject. Group terms are differences from controls. "All groups equal" is a Wald test of the three group terms. $p < .05$ in bold.

	Alpha		Beta		Gamma	
Term	Estimate (SE)	p	Estimate (SE)	p	Estimate (SE)	p
MED − IMW (first block)	−0.005 (0.026)	.851	−0.013 (0.020)	.513	−0.008 (0.034)	.824
think2 − think1	+0.024 (0.029)	.412	+0.009 (0.025)	.717	−0.000 (0.033)	.995
(med2 − med1) − (think2 − think1)	+0.015 (0.031)	.627	+0.052 (0.036)	.148	+0.002 (0.046)	.971
med2 − med1breath	+0.039 (0.020)	.055	+0.061 (0.023)	**.009**	+0.001 (0.036)	.968
Second pair	−0.038 (0.023)	.089	−0.043 (0.018)	**.014**	−0.041 (0.023)	.069
Observations (subjects)	117 (40)		117 (40)		117 (40)	

Table 6: H_2: OLS of the DFA exponent for meditators on task (MED or IMW), position in the pair (first or second block), their interaction and the second-pair indicator, with standard errors clustered by subject. The first three rows are model terms; "med2 − med1breath" is the sum of the second and third. $p < .05$ in bold.